

***Operating System***

***Lab Project***

***COMP-262L***



***Bachelors of Artificial Intelligence***

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# CPU Scheduling Simulator using Python

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## **1. Abstract**

This project presents a CPU Scheduling Simulator developed in Python to demonstrate core concepts of process scheduling in operating systems. The simulator implements four commonly used CPU scheduling algorithms: First Come First Serve (FCFS), Shortest Job First (SJF), Priority Scheduling, and Round Robin. The system accepts process details such as Process ID, Arrival Time, Burst Time, and Priority, then computes Waiting Time and Turnaround Time. It also generates a Gantt Chart for visualization of process execution.

## **2. Introduction**

CPU scheduling is one of the fundamental responsibilities of an operating system. It determines which process gets CPU time and for how long. Efficient scheduling improves CPU utilization, minimizes waiting time, and enhances system performance. This project provides a practical simulation environment for understanding scheduling techniques.

## **3. Objectives**

- To simulate CPU scheduling algorithms.
- To calculate Waiting Time and Turnaround Time.
- To visualize scheduling using Gantt Charts.
- To compare algorithm performance.
- To build a graphical interface for easy interaction.

## **4. Technologies Used**

- **Programming Language:** Python
- **GUI Library:** Tkinter
- **Visualization Library:** Matplotlib
- **Development Environment:** Visual Studio Code

## **5. Scheduling Algorithms Implemented**

### **1. First Come First Serve (FCFS)**

FCFS executes processes in the order they arrive. It is simple to implement but may cause long waiting times for shorter processes.

### **2. Shortest Job First (SJF)**

SJF selects the process with the shortest burst time. It minimizes average waiting time but may cause starvation for long processes.

### **3. Priority Scheduling**

Each process is assigned a priority. Processes with higher priority are executed first.

### **4. Round Robin**

Round Robin allocates CPU time in fixed time slices called quantum, improving fairness and responsiveness.

## **6. System Design**

The project consists of three modules:

1. **Algorithm Module** – Implements scheduling logic.
2. **GUI Module** – Handles user interaction.
3. **Visualization Module** – Displays Gantt Charts.

## **7. Formula Used**

**Turnaround Time = Completion Time – Arrival Time**

**Waiting Time = Turnaround Time – Burst Time**

### **Features**

- Add custom processes
- Select scheduling algorithm
- Enter time quantum for Round Robin
- View scheduling results

- Display Gantt Chart
- Compare scheduling behavior

## **8. Results**

The simulator successfully executes all four scheduling algorithms and visually demonstrates process execution. It provides accurate calculation of Waiting Time and Turnaround Time, helping students understand CPU scheduling concepts practically.

### **Advantages**

- Easy to use
- Educational and interactive
- Fast simulation
- Clear visualization
- Useful for Operating Systems learning

### **Limitations**

- Supports basic scheduling only
- No multi-core CPU simulation
- Does not include advanced real-time scheduling

### **Future Improvements**

- Add advanced scheduling algorithms
- Add export to CSV/PDF
- Include comparison graphs
- Improve UI design
- Add real-time simulation mode

## **9. Conclusion**

The CPU Scheduling Simulator is a practical implementation of Operating Systems concepts. It demonstrates how different scheduling algorithms affect CPU performance, waiting time, and turnaround time. The project improves conceptual understanding through simulation and visualization, making it useful for academic learning.